

Data Request and Requirements Guide

v0.1

> PRE-KICKOFF METHOD DRAFT。此清單描述 analysis needs，不代表 AMAT 現有資料或已承諾提供。
資料處理須符合 NDA、InfoSec 與 client authorization。

1. Start from the decision

資料不是越多越好。每一項 request 必須連到明確 decision / model use：

Decision / output	Minimum viable input	Expansion trigger
1.4 annual residual gap	site / entity master、annual or monthly consumption、forecast、target boundary、existing eligible procurement	需要 seasonality、site / meter allocation、hourly matching 或供應 profile 時再取 interval data
1.5 portfolio scenarios	residual gap、option eligibility、timing、existing tenor / expiry、cost baseline / price components、constraints	需要 finance approval、budget volatility、hourly profile 或 technology matching 時擴充
1.6 roadmap	decision deadlines、stakeholder / approval lead time、contract expiry、data / implementation dependencies	有 external deadline、site expansion / closure、system change 時擴充
2.x market engagement	approved requirement band、sites / delivery points、technology / tenor / flexibility、disclosure authorization	named RFQ 前才提供更細需求，且所有 bidder 接收一致 material information
3.x negotiation / execution	authorized commercial assumptions、legal / procurement process、implementation contacts、meter / certificate setup	only after shortlist / mandate and need-to-know approval

2. Data request catalogue

A. Boundary and master data

Data group	Minimum fields	Why needed	Truth status
Entity / site master	stable site ID 、 entity 、 site name / pseudonym 、 location level needed 、 status 、 open / close date 、 included-in-target flag	prevent missing / duplicate sites and map approvals	TBD WITH CLIENT
Meter / account mapping	stable meter / account ID 、 site ID 、 utility / service reference 、 valid-from / to	reconcile consumption and future implementation	TBD WITH CLIENT
Target / policy	target metric 、 target years 、 scope boundary 、 eligible instruments / evidence 、 ownership / retirement requirements	determine numerator / denominator and option eligibility	TBD WITH CLIENT; formal policy source required

B. Electricity demand

Data group	Minimum fields	Quality expectation	Downstream use
Historical consumption	site / meter key 、 period start / end 、 MWh / kWh 、 actual / estimate 、 gross / net flag 、 source system	unique key; unit known; totals reconcile to owner report	baseline 、 seasonality 、 gap
Forecast	site / entity 、 period 、 value 、 unit 、 scenario 、 version 、 forecast date 、 owner 、 driver	version controlled; drivers / confidence explained	target-year gap 、 portfolio volume
Site changes	expansion / closure / efficiency event 、 date 、 volume impact range 、 confidence 、 owner	no single-point value if material uncertainty	sensitivity 、 roadmap

C. Existing procurement and certificates

Data group	Minimum fields	Quality expectation	Downstream use
Contract / program summary	instrument ID 、 counterparty / internal source 、 site allocation 、 start / end 、 committed / expected MWh 、 tenor 、 technology 、 delivery structure	term dates and allocation must not overlap without explanation	existing coverage / expiry
Certificate / claim	instrument ID 、 certificate type 、 vintage / generation period 、 ownership / retirement status 、 beneficiary 、 eligible amount 、 evidence location	certificate amount / period reconciles to contract or registry record	claim / gap treatment
Onsite generation	site 、 asset ID 、 technology 、 capacity 、 generation actual / forecast 、 self-use / export 、 certificate treatment	actual vs forecast separated	baseline / portfolio

D. Commercial and implementation context

Data group	Minimum fields	Handling note	Downstream use
Electricity cost baseline	billing period 、 site 、 energy / demand / wheeling / fee / tax components 、 unit 、 currency	Client Confidential ; 只取 scenario decision 所需	1.5 comparison
Existing contract terms	price structure 、 indexation 、 volume / flexibility 、 termination / change 、 credit 、 guarantee 、 settlement 、 certificate terms	Highly Restricted by default ; need-to-know	1.5 / 3.x

Data group	Minimum fields	Handling note	Downstream use
Procurement / approval process	approval gates 、 thresholds 、 lead time 、 required functions 、 vendor onboarding	role names / limits TBD	roadmap / RFQ / signing
Budget / risk parameters	budget cycle 、 evaluation horizon 、 discount / escalation assumptions 、 risk tolerances	client-authorized only	portfolio / negotiation

3. Required metadata per file

- Request ID 、 file owner 、 business owner 、 source system 。
- Classification 、 permitted purpose 、 permitted recipients 、 retention / deletion rule 。
- Extract date 、 coverage period 、 timezone 、 calendar / fiscal convention 。
- Grain / primary key 、 units 、 currency 、 actual / estimate / forecast flag 。
- Transformations / filters 、 version 、 known limitations 、 reconciliation total 。

4. Data completeness and quality checklist

Test	Question	Example control	Decision if failed
Completeness	Required sites / periods / fields present?	compare site master and period matrix	request missing; range; exclude; re-scope
Validity	Types 、 units 、 dates 、 codes valid?	controlled dictionary / allowed values	correct mapping or reject row
Uniqueness	Key duplicated?	site-meter-period unique check	explain aggregation / deduplicate with evidence
Consistency	Gross / net 、 actual / forecast 、 timezone 、 unit consistent?	standardize and retain raw field	document transform; sensitivity if material
Reconciliation	Totals tie to owner report / invoice / certificate evidence?	variance amount / % with threshold	investigate; condition G1 readiness
Lineage	Can each number trace to file / row / transform?	source ID + version + mapping log	do not use as decision evidence

Test	Question	Example control	Decision if failed
Currency / price	Cost components and FX / escalation treatment explicit?	component-level dictionary	no single blended price assumption
Confidentiality	Is every field necessary and authorized?	classification / recipient / tool review	minimize, anonymise or stop

Quality status 建議使用 Pass / Pass with condition / Fail / Not tested ; threshold 需由 client / model owner 確認，不在 pre-work 中發明。

5. Stakeholder interview guide

Sustainability / RE100 owner

- 哪個 target、boundary、reporting period、eligible evidence 與 claim rule 由哪份正式 policy 定義？
- 如何處理 acquisition / divestment、新 site、onsite generation、unbundled certificates、contracted but not yet operational supply？
- 哪些結果要給 Taiwan / regional / global decision body？

Facilities / site energy owner

- Source system、meter / account mapping、actual / estimate、gross / net、timezone / billing-cycle 差異為何？
- 哪些 site 有 expansion / closure / efficiency changes？哪些數值是 approved forecast？
- Wheeling / implementation 需要哪些 account / meter / contact 資料，何時才應取得？

Procurement

- Existing contracts、supplier / vendor onboarding、RFQ rules、approval thresholds、timeline、required comparability fields？
- 哪些資料可在 anonymous sounding、named outreach、RFQ、negotiation 分別揭露？

Finance

- Cost baseline 應採哪些 components、currency、tax、discount / escalation assumptions 與 reporting horizon？
- 哪種 cost / risk output 可支持 budget / approval？

Legal / InfoSec

- NDA、data transfer、retention、external advisor / tool、AI / SaaS、privilege 與 personal data rules ?
- Commercial review 與 legal advice 如何分工 ?

6. Data gap decision process

1. Log : 記錄 missing / unreliable field、affected rows / period、source owner。
2. Size : 估算 impact range，不把「資料缺」直接等同低重要性。
3. Choose treatment : Resolve、Use approved assumption、Use range / sensitivity、Exclude、Re-scope。
4. Approve : material treatment 由 business / decision owner 確認；不是 analyst 自行隱含決定。
5. Trace : gap ID 連到 model assumption、output caveat 與 decision record。
6. Close / carry : 取得 evidence 後關閉，或在 gate 明確標示 condition。

7. G1 data readiness decision

Ready

Minimum fields 可追溯、定義一致、material reconciliation 通過、assumptions 已批准，可支持目前 decision。

Ready with conditions

仍有 gaps，但 impact 已量化、sensitivity / range 足以決策、owner / due date / condition 已記錄。

Not ready

Boundary / target / primary key / material volume / existing procurement 無法確認，結果可能反轉 recommendation；需要補資料或 re-scope。

8. Requirement collection questions

Question	Why it matters	Needed before
Client 希望 gap / portfolio output 支持哪個 approval 或 budget cycle ?	決定 horizon、grain、format	data request launch
Baseline 是 calendar year、fiscal year 或 rolling period ?	避免 period mismatch	1.4 mapping

Question	Why it matters	Needed before
Target boundary 包含哪些 entities / sites / loads ?	denominator	G1
Existing procurement 的 eligibility / allocation 由誰批准 ?	numerator / claim	G1
Forecast 的 approved version 與 change process ?	target-year volume	G2
哪些缺口可以 assumption / range 前進，誰批准 ?	material uncertainty control	G1
是否真的需要 hourly / monthly granularity ?	防止無用途的 data burden	before expansion

9. First client-facing output

不是把所有 raw data 退回 client，而是一頁 readiness summary：coverage by required category、top material gaps、treatment / owner / due、decision impacts、G1 recommendation。Workbook 的 Completeness sheet 已提供可重用格式。